Singular Value Decomposition (SVD)

↳ So far we have seen the following decompositions:

$A = CR$ (Column and Row Matrix in 1.4)

$A = LU$ (lower and upper triangular Matrix in 2.3)

$A = QR$ (Orthonormal matrix in 4.4) not in syllabus

$A = S\Lambda S^{-1}$ (Eigenmatrix in 6.2)

Now, here comes the grand finale of linear algebra:

$$\boxed{A = U\Sigma V^T}$$

Before we dive in let us go through some basic stuff:

So far we have done the decomposition of square matrices in most cases. But for **SVD** decomposition, it does not have to be a square matrix. It can be applied for any matrix.

For now, imagine two vector:

$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\mathbb{R}^2$ 2×1

$B = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ $\mathbb{R}^3$ 3×1

They look in same but in different dimension.
But how can we transform from A to B or B to A?

So, to transform A to B, we need a 3×2 matrix C.

So, $B = CA$.

Again, to transform B to A, we need a 2×3 matrix D.

So, $A = DB$.

Rectangular matrix!

Here both C and D are rectangular matrix.

So, a matrix of m*n has the power to transform a vector from n-th dimension to m-th dimension. Just basic linear transformation.

Now think of a simple case first. What to do if you want to turn a 3D vector to 2D vector. You linearly transform it with this vector:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(Dimension Eraser)

this completely removes the z value from 3D vector

Now how about 2D vector to 3D vector:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

this adds a dimension and keeps the value

(Dimension Adder) as 0.

We have talked about rectangular matrices and how it alters the dimension of a matrix. Now let us Recap a bit of Symmetric Matrix.

- $A^T = A$
- If A is symmetric and square,
  - their eigenvalues are real
  - their eigenvectors are orthogonal
- If A is rectangular (which is of course not symmetric):

$B = AA^T$, B is symmetric and square.

$$\underset{A \;\; 2\times3}{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}} \underset{A^T \;\; 3\times2}{\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}} = \underset{B \;\; 2\times2}{\begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}}$$

Also, $C = A^TA$, C is also square and symmetrical.

$$C = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

$$S_L = \begin{bmatrix} | & | \\ \vec{u_1} & \vec{u_2} \\ | & | \end{bmatrix} \rightarrow \lambda_1 \; \lambda_2$$

$\lambda_1 \geq \lambda_2$

$$\left.\begin{matrix} 27 & 36 & 45 \end{matrix}\right]$$

For now, let's call $AA^T\ (B) \rightarrow S_L$

$A^TA\ (C) \rightarrow S_R$

and

$\lambda_1 \geq \lambda_2$

$$S_R = \begin{bmatrix} | & | & | \\ \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \\ | & | & | \end{bmatrix} \rightarrow \lambda_1\ \lambda_2\ \lambda_3$$

$\lambda_1 \geq \lambda_2 \geq \lambda_3$

So, for this case,

$S_L \rightarrow 2$ eigenvectors $\rightarrow$ left singular vectors of A

$S_R \rightarrow 3$ eigenvectors $\rightarrow$ right singular vectors of A

Here S_L and S_R are Positive Semi-Definite Matrices.

This implies that eigenvalue for each eigenvectors are non-negative. ($\lambda_i \geq 0$)

$\hookrightarrow$ Biggest eigenvalue of S_L is equal to the biggest eigenvalue of S_R.

Same for smallest one.

The leftover eigenvalue for S_R is guaranteed to be 0.

Assume the equal values are λ_1 and λ_2.

So, $\left.\begin{array}{l}\sqrt{\lambda_1} = \sigma_1 \\ \sqrt{\lambda_2} = \sigma_2\end{array}\right\}$ Singular values of A

We are done with the basic part.

Time for the entrance of SVD.

Any matrix A can be decomposed into three special matrix:

$$A = U \Sigma V^T$$

Here, $U_{m \times m}$ → orthonormal matrix

→ rotate standard basis to left singular vectors

→ Contains normalized eigen vectors S_i

* $\Sigma_{m \times n}$ → Diagonal matrix with singular values

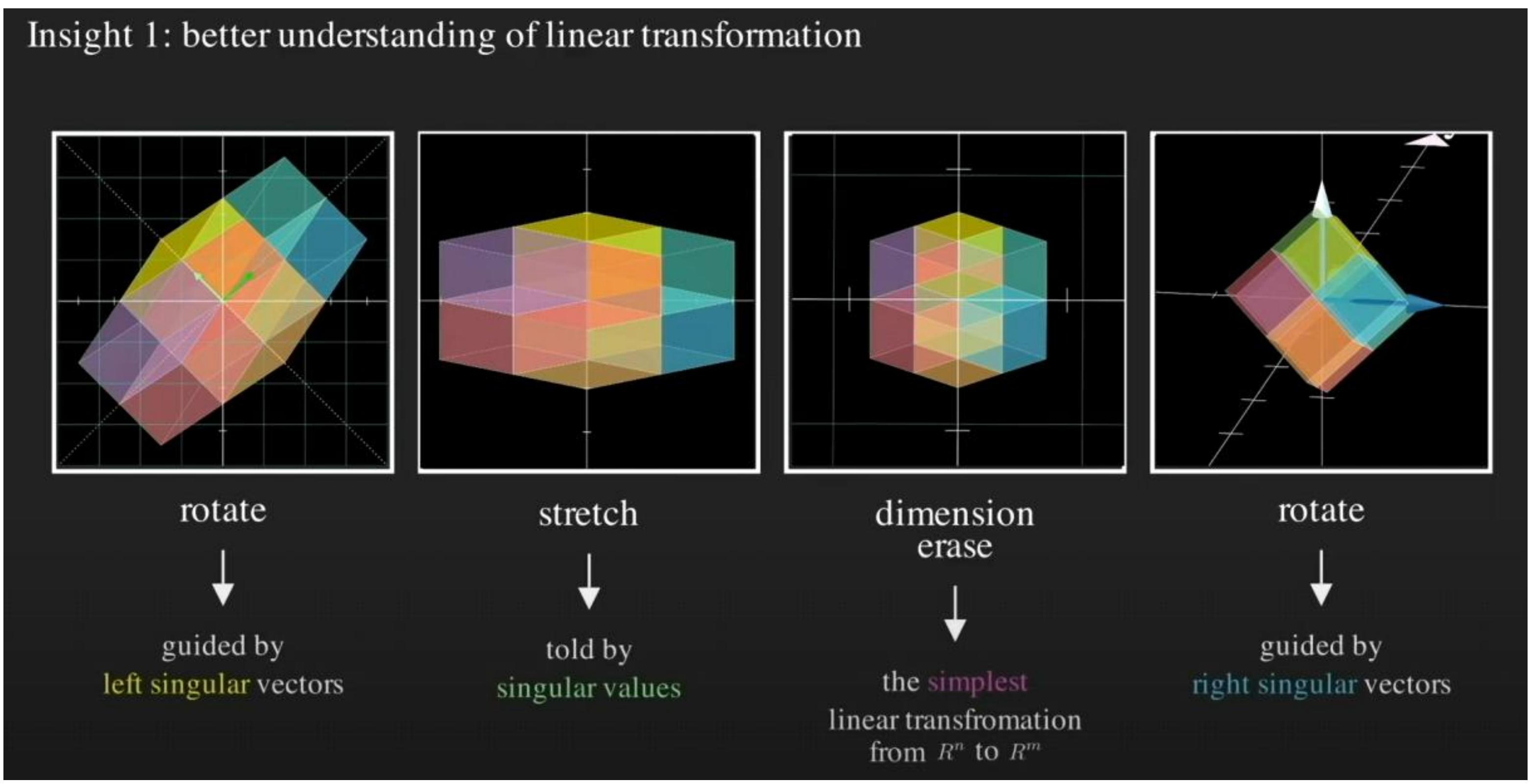
→ Removes or adds dimension and scales or shrink axis

→ Singular values will be in descending order

* $V_{n \times n}$ → orthonormal matrix

→ rotate right singular vectors to standard basis

→ Contains normalized eigenvectors of S_R



Example 1:

$$A = \begin{bmatrix} 5 & 4 \\ 0 & 3 \end{bmatrix}$$

Step 1: Find their S_L and S_R.

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$$S_L = AA^T = \begin{bmatrix} 41 & 12 \\ 12 & 9 \end{bmatrix}$$

$$S_R = A^T A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

Step 2: Find the normalized eigenvectors of S_L and S_R

For S_L, $\det(S_L - \lambda I) = 0$

$$\Rightarrow \begin{vmatrix} -\lambda & 12 \\ 12 & 9-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (41-\lambda)(9-\lambda) - 144 = 0$$

$$\Rightarrow 369 - 50\lambda + \lambda^2 - 144 = 0$$

$$\Rightarrow \lambda^2 - 50\lambda + 225 = 0$$

$$\Rightarrow \lambda = 5, 45$$

Now, $(S_L - \lambda I)x = 0$ (Always start with the larger λ)

$\lambda = 45$

$\hookrightarrow \begin{bmatrix} -4 & 12 \\ 12 & -36 \end{bmatrix} \begin{bmatrix} x \\ x_2 \end{bmatrix} = 0$

$\hookrightarrow u_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$\therefore u_{1n} = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$

$\lambda = 5$

$\hookrightarrow \begin{bmatrix} 36 & 12 \\ 12 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\hookrightarrow u_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

$\therefore u_{2n} = \begin{vmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{vmatrix}$

For S_R, $\det(S_R - \lambda I) = 0$

$\Rightarrow \begin{vmatrix} 25-\lambda & 20 \\ 20 & 25-\lambda \end{vmatrix} = 0$

$\Rightarrow (25-\lambda)^2 - 20^2 = 0$

$\Rightarrow (45-\lambda)(5-\lambda) = 0$

$\Rightarrow \lambda = 5, 45$ (the eigenvalues of S_L and S_R would be same)

Now, $(S_R - \lambda I)x = 0$

$\hookrightarrow$ $\lambda = 45$

$\lambda = 5$ $\begin{bmatrix} 20 & 20 \end{bmatrix} \begin{bmatrix} x_1 \end{bmatrix} = 0$

$\lambda = 45$

$\hookrightarrow \begin{bmatrix} -20 & 20 \\ 20 & -20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \; 0$

$\hookrightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hookrightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore v_{1n} \quad \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$\lambda = 5$

$\hookrightarrow \begin{bmatrix} 20 & 20 \\ 20 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\hookrightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\hookrightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore v_{2n} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

Step 3: Find Σ

(lol $\Sigma\Sigma$ boy Σ boy)

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} = \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

Remember, $\sigma_1 \geq \sigma_2 \rightsquigarrow \lambda_1 \geq \lambda_2$

Step 4: Finally Place U, V, Σ

$$U = \begin{bmatrix} u_{1n} & u_{2n} \end{bmatrix} = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix}$$

$$V = \begin{bmatrix} v_{1n} & v_{2n} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$\therefore A = U \Sigma V^T$$

$$= \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \begin{bmatrix} 2\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$$


done!